Fig 14 — Transfer to Tetouan without re-tuning: what is frozen, what is learned from unlabelled data, what one city changes

SOURCE · Morocco (UCI 1158)
15 zones · 3 cities · 2022–24 · A

Chosen here, then frozen:
• 19 rules and all thresholds
• label-model π, θ (Dawid–Skene)
• detector design + training config
• event formation (w, q, gap)
on development seeds 100–104

frozen

PIPELINE STAGE	FROZEN from Morocco	RE-ESTIMATED from Tetouan's unlabelled series	ONE-CITY ADAPTATION (structural, not tuned)
1 Per-step states → items	dead ε = 5 % of x̂, hold ≥ 6, linear fill ≥ 6 points (relative: kW vs A needs no change)	x̂ = same-slot median of the previous 14 days	—
2 19 labelling rules vote	every threshold (τ, η, r, π, κ, δ*, hours, durations)	—	R1 multi-city zero and R7 regional outage cannot fire
3 Label model	π, θ fitted on Morocco; applied by one E-step (no refit); 0 unseen rule outputs	—	—
4 Predictor: TCN-GAT	architecture + training config (Huber, AdamW, 40 epochs, patience 6, seed 0)	weights, on rule-clean steps; 2 folds split at 1 Jul 2017	graph = 3 same-city edges, no cross-city edge
5 Multi-scale residual hierarchy	MODWT LA(8) J = 4, phase alignment, robust z (28-day window)	score scales	city part ≡ region part (u = 0)
6 Event formation	smoothing w = 24, percentile q = 99.9, merge gap = 12, run ≥ 3	γ = 99.9th percentile of Tetouan's rule-negative steps	—

TARGET · Tetouan (UCI)
3 zones · 1 city · 2017 · kW
no labels · pre-cleaned series

run the frozen pipeline

REAL CLEAN YEAR

0 false rule items
(only the 4 clock days, all N)

7 detector alarms in 36 zone-months

2,778 INJECTED EVENTS (30 datasets)

event F1, rules + detector
0.945 · 0.919 · 0.890
at 1 · 3 · 7 %

precision 0.98–1.00
above rules alone (0.856 · 0.861 · 0.860)
in 10/10 seeds at every rate

ATTRIBUTION (frozen label model)

5 of 6 classes: 0.85–1.00 accuracy
macro-F1 without A1: 0.95–0.98

A1 → G2: zero gap = city outage
when there is only one city

ENGINEERING ONLY (no threshold or setting changed): rule engine accepts a series with no anomaly (3 empty-table fixes) · label model predict() · single-city injector (G1 not injected; A1 and U_SHOCK city-wide; A2 2–3 zones or city-wide) · cross-fitting split read at run time; graph top-k clamped · 2017 calendar and clock dates · item schema aligned. Morocco reproduced exactly after every change (catalogue, label model, injector, TCN-GAT MAE 0.0294). No file changed after the first injected Tetouan result (docs/transfer_protocol.md §5).

- frozen from Morocco
- re-estimated from unlabelled target data
- single-city adaptation
- result on Tetouan